

Medvale Internal Medicine

Chapter 1A: Approach to Chest Pain

Original study resource draft

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Contents

1 Chapter 1A: Approach to Chest Pain

1.1 Learning objectives

By the end of this chapter, the learner should be able to:

1. Identify unstable chest-pain patients who need immediate resuscitation.
2. Build a life-threatening differential before settling on a benign diagnosis.
3. Use ECG and serial troponin testing as decision tools rather than isolated yes/no tests.
4. Distinguish stable angina, unstable angina, NSTEMI, STEMI, pericarditis, pulmonary embolism, aortic dissection, pneumothorax, tamponade, and esophageal rupture by pattern recognition.
5. Decide which patients need ED/inpatient management, observation, urgent imaging, reperfusion, or outpatient follow-up.

1.2 Why this matters clinically

Chest pain is one of the highest-stakes complaints in internal medicine because the first decision is not, “What is the final diagnosis?” The first decision is, “Could this patient die soon if I miss the diagnosis?” A patient with reflux, costochondritis, panic symptoms, acute coronary syndrome, aortic dissection, pulmonary embolism, pericarditis, and tension pneumothorax may all initially say the same sentence: “My chest hurts.”

The safe approach is pattern-based and time-sensitive. In acute chest pain, the clinician screens first for unstable physiology and life-threatening disease, then uses ECG, troponin pattern, vital signs, focused history, physical examination, and targeted imaging to separate emergencies from lower-risk outpatient problems.

The most important exam habit is to avoid premature closure. Chest pain that sounds atypical can still be dangerous, and chest pain that sounds classic still requires proof and localization.

1.3 The first question: sick or not sick?

Before asking whether the pain is “typical,” decide whether the patient is unstable.

1.3.1 Immediate danger signs

Danger sign	Why it matters
Hypotension	Shock from MI, dissection, PE, tamponade, or tension pneumothorax
Hypoxemia	PE, pneumothorax, pulmonary edema, pneumonia, severe ACS
Altered mental status	Poor perfusion, hypoxemia, severe shock
Syncope	Dissection, massive PE, malignant arrhythmia, severe aortic stenosis
New neurologic deficit	Aortic dissection involving great vessels or embolic event
Severe tearing pain radiating to the back	Aortic dissection until proven otherwise
Unilateral absent breath sounds with instability	Tension pneumothorax
JVD with hypotension	Tamponade, tension pneumothorax, RV MI, massive PE
New murmur after chest pain	Papillary muscle rupture, ventricular septal rupture, acute valve catastrophe
Crushing pressure with diaphoresis	ACS or MI until proven otherwise

Medvale rule: If the patient looks unstable, do not wait for the complete differential. Stabilize airway, breathing, and circulation while obtaining ECG, IV access, cardiac monitoring, oxygen if hypoxemic, and urgent bedside evaluation.

1.4 Initial diagnostic bundle

For most adults with acute chest pain, the first-pass evaluation includes vital signs, pulse oximetry, ECG, cardiac biomarkers, and often chest radiography. Unstable patients may also need bedside echocardiography or point-of-care ultrasound.

Step	Test or action	Main purpose
1	Vitals and oxygen saturation	Determine stability
2	Focused history	Identify ACS, PE, dissection, pneumothorax, pericarditis, GI, and MSK patterns
3	12-lead ECG	Look for STEMI, ischemia, arrhythmia, pericarditis clues
4	Troponin, preferably serial	Detect myocardial injury and trend over time
5	Chest x-ray	Pneumothorax, widened mediastinum, pneumonia, pulmonary edema
6	Basic labs	Anemia, renal function, electrolytes, medication safety
7	Targeted imaging	CTA chest, CT pulmonary angiography, echo, or other testing based on suspected diagnosis

A normal first ECG or first troponin does not automatically exclude acute coronary syndrome. The story, symptom timing, serial ECGs, and serial troponins matter.

1.5 Core illness scripts

1.5.1 Acute coronary syndrome

Core idea: plaque disruption, thrombosis, or supply-demand imbalance causes acute myocardial ischemia. **Typical pain:** pressure, heaviness, squeezing, or burning discomfort, often substernal, sometimes radiating to arm, jaw, neck, back, or epigastrium. **Associated clues:** diaphoresis, nausea, dyspnea, syncope, new heart failure, ischemic ECG changes, troponin rise. **Do first:** ECG, serial troponins, aspirin when not contraindicated, cardiac monitoring, risk stratification, and reperfusion pathway if STEMI.

1.5.2 Aortic dissection

Core idea: an intimal tear allows blood to split layers of the aortic wall. **Typical pain:** sudden maximal-onset severe chest or back pain, often described as tearing or ripping. **Associated clues:** pulse deficit, neurologic deficit, syncope, aortic regurgitation murmur, mediastinal widening, severe hypertension or shock. **Do first:** control heart rate and blood pressure if stable, urgent CTA if stable, TEE/bedside evaluation if unstable, surgical consultation for ascending dissection.

1.5.3 Pulmonary embolism

Core idea: thrombus obstructs pulmonary arterial flow and can strain the right heart. **Typical pain:** pleuritic chest pain with dyspnea. **Associated clues:** tachycardia, hypoxemia, hemoptysis, unilateral leg swelling, recent surgery, immobility, malignancy, estrogen exposure, prior VTE. **Do first:** estimate pretest probability, then use PERC/D-dimer/CT pulmonary angiography as appropriate; anticoagulate if sufficiently likely and safe.

1.5.4 Tension pneumothorax

Core idea: air trapped in the pleural space causes lung collapse and obstructive shock. **Typical pain:** sudden unilateral pleuritic chest pain and dyspnea. **Associated clues:** hypotension, unilateral absent breath sounds, JVD, tracheal deviation late. **Do first:** if unstable, decompress immediately; do not wait for imaging.

1.5.5 Pericarditis and myopericarditis

Core idea: inflamed pericardium causes pleuritic and positional pain; myocardium may also be involved. **Typical pain:** sharp or pleuritic pain, worse lying back, better leaning forward. **Associated clues:** pericardial friction rub, diffuse ST elevation, PR depression, possible troponin elevation if myocarditis is present. **Do first:** ECG, troponin, inflammatory markers, echo if effusion or tamponade concern.

1.5.6 Esophageal rupture

Core idea: full-thickness esophageal tear after forceful vomiting contaminates the mediastinum. **Typical pain:** severe chest pain after vomiting or instrumentation. **Associated clues:** fever, sepsis, subcutaneous emphysema, pleural effusion. **Do first:** NPO, broad-spectrum antibiotics, urgent imaging and surgical consultation.

1.6 Chest pain pattern recognition

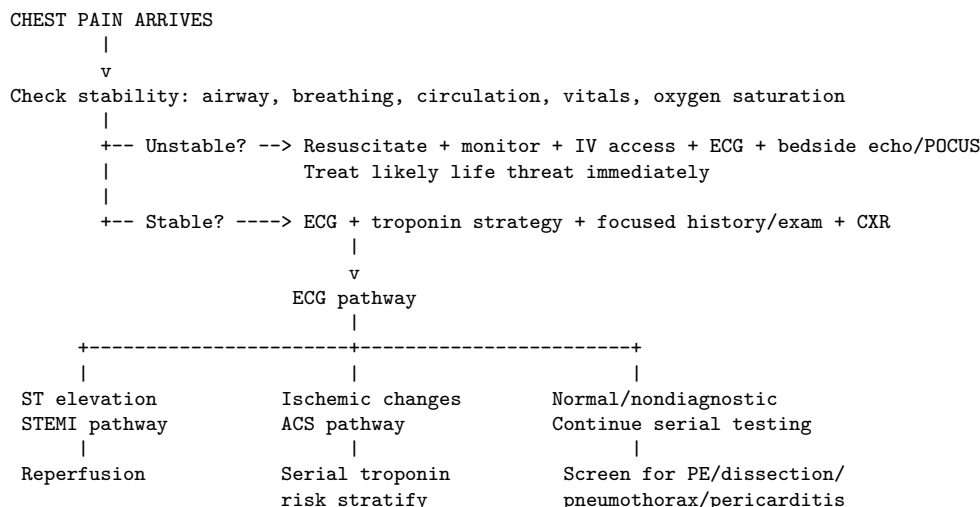
Chest pain history is useful, but it is not perfect. The goal is not to memorize one classic presentation. The goal is to decide which dangerous diagnosis must be ruled out first.

Pain pattern	Diagnosis to think about	Important caution
Pressure, heaviness, squeezing, substernal discomfort	ACS, stable angina	Older patients, women, and diabetic patients may have atypical symptoms
Pain with exertion, relieved by rest	Stable angina	New, worsening, or rest pain becomes unstable angina/ACS
Sudden tearing pain radiating to back	Aortic dissection	Check pulse deficits, neurologic symptoms, BP difference
Pleuritic pain with dyspnea or tachycardia	PE, pneumothorax, pericarditis, pneumonia	Pleuritic does not mean benign
Positional pain better leaning forward	Pericarditis	Still obtain ECG and troponin
Reproducible chest wall tenderness	Costochondritis/MSK pain	Reproducible pain lowers concern but does not fully exclude ACS
Burning after meals, sour taste	GERD/esophageal spasm	Esophageal pain can mimic angina
Severe vomiting then chest pain	Esophageal rupture	Look for sepsis, crepitus, pleural effusion
Chest pain with panic, paresthesias, hyperventilation	Panic attack	Diagnose after dangerous causes are reasonably excluded

1.7 Life-threatening chest pain comparison table

Diagnosis	Typical clues	Initial test	Immediate management priority
ACS / MI	Pressure, diaphoresis, dyspnea, risk factors, ECG changes, troponin rise	ECG + serial troponin	Antiplatelet/anticoagulation strategy, anti-ischemic therapy, reperfusion if STEMI
Aortic dissection	Sudden tearing pain to back, pulse/BP differential, neuro deficit, widened mediastinum	CTA chest if stable; TEE/bedside echo if unstable	BP/HR control, urgent surgery for ascending dissection
Pulmonary embolism	Pleuritic pain, dyspnea, tachycardia, hypoxemia, DVT risk	Risk score, then D-dimer or CT pulmonary angiography	Anticoagulation or thrombolysis/embolectomy if massive PE
Tension pneumothorax	Sudden dyspnea, unilateral decreased breath sounds, hypotension, JVD	Clinical diagnosis if unstable	Immediate needle decompression/chest tube
Cardiac tamponade	Hypotension, JVD, muffled heart sounds, pulsus paradoxus	Bedside echo	Pericardiocentesis if unstable
Esophageal rupture	Severe chest pain after vomiting, sepsis, subcutaneous emphysema	Contrast esophagram or CT	Broad antibiotics, NPO, surgical consult

1.8 Original diagram 1: Acute chest pain first-pass triage



1.9 ECG-first reasoning

The ECG is not just a test for myocardial infarction. It is a sorting tool.

ECG finding	Think about
ST elevation in an anatomic coronary territory	STEMI
ST depression or T-wave inversion in a concerning story	NSTEMI/unstable angina
New left bundle branch block with ischemic symptoms	Possible acute coronary occlusion; interpret carefully
Diffuse concave ST elevation + PR depression	Acute pericarditis
Electrical alternans or low voltage	Large pericardial effusion/tamponade
Right heart strain pattern	PE; not sensitive enough to rule in/out alone
Wide-complex tachycardia	Ventricular tachycardia until proven otherwise
Bradycardia or AV block with inferior MI symptoms	RCA/inferior MI involvement

Exam clue: ECG localization matters most when it changes management. ST elevation in II, III, and aVF suggests inferior MI. ST elevation in V1-V4 suggests septal/anterior involvement. ST elevation in I, aVL, V5, and V6 suggests lateral involvement. Posterior MI may show ST depression in V1-V3 with tall R waves and should be confirmed with posterior leads.

1.10 Troponin-first reasoning

Troponin identifies myocardial injury, not automatically type 1 plaque-rupture MI. Troponin can rise in ACS, demand ischemia, myocarditis, heart failure, renal disease, sepsis, pulmonary embolism, tachyarrhythmia, or cardiac contusion.

Troponin pattern	Interpretation
Normal initially, symptoms just started	Repeat; early MI can be missed
Rising/falling pattern with ischemic symptoms	Supports acute MI
Elevated but flat/chronic	Chronic myocardial injury or renal disease may be present
Elevated with sepsis, anemia, tachycardia, hypoxia	Consider demand ischemia/type 2 MI
Elevated with pleuritic/positional pain and viral symptoms	Consider myopericarditis
Elevated with PE signs	RV strain from PE can injure myocardium

Medvale rule: Troponin answers, “Is there myocardial injury?” The clinical picture and ECG answer, “Why?”

1.11 ACS versus stable angina versus noncardiac pain

1.11.1 Stable angina

Stable angina is predictable ischemic discomfort caused by myocardial oxygen demand exceeding supply, usually in the setting of fixed coronary atherosclerosis. It is typically provoked by exertion or emotional stress and improves with rest or nitroglycerin. The uploaded cardiology excerpt emphasizes this classic pattern and then moves into stress testing, catheterization, and medical management.

1.11.2 Unstable angina / NSTEMI

Unstable angina and NSTEMI are acute coronary syndromes. They are suggested by chest pain that is new, worsening, occurring at rest, lasting longer, or requiring less exertion to trigger. NSTEMI has myocardial necrosis and therefore troponin elevation; unstable angina does not.

1.11.3 STEMI

STEMI is an ECG diagnosis requiring urgent reperfusion. Do not wait for troponin to become positive if the ECG and clinical picture show STEMI.

1.11.4 Noncardiac pain

Noncardiac pain is considered only after dangerous cardiac, vascular, and pulmonary causes have been addressed. GERD, esophageal spasm, costochondritis, pneumonia, anxiety/panic, and herpes zoster are common mimics.

1.12 Diagnostic approach by risk level

1.12.1 High-risk chest pain

High-risk patients usually need ED or inpatient management.

Feature	Why it raises risk
Ischemic ECG changes	Possible ACS
Positive or rising troponin	Myocardial injury
Persistent pain despite therapy	Ongoing ischemia or alternative catastrophe
Hemodynamic instability	Shock physiology
Ventricular arrhythmia	Electrical instability
New heart failure	Large MI, valve rupture, myocarditis, severe cardiomyopathy
Syncope with chest pain	PE, dissection, arrhythmia, severe outflow obstruction
Known CAD with typical symptoms	Higher pretest probability

1.12.2 Intermediate-risk chest pain

Intermediate-risk patients are stable but not clearly safe. They often need observation, serial ECG/troponin testing, and further testing such as stress imaging or coronary CT angiography depending on the scenario and local practice.

1.12.3 Low-risk chest pain

Low-risk patients have reassuring history, normal ECG, negative serial troponins when appropriately timed, stable vitals, and no concerning alternative diagnosis. They may be discharged with follow-up and risk-factor management when clinically appropriate.

1.13 Common mimics and how to separate them

Mimic	Clues	What not to miss
GERD	Burning, postprandial, worse lying down, sour taste	Inferior MI can feel epigastric
Esophageal spasm	Squeezing pain, may respond to nitrates	Can be indistinguishable from angina by symptoms alone
Costochondritis	Localized reproducible tenderness	ACS can coexist with chest wall tenderness
Panic attack	Fear, palpitations, paresthesias, hyperventilation	PE, arrhythmia, ACS
Pneumonia	Fever, cough, pleuritic pain, infiltrate	PE or ACS with incidental cough
Herpes zoster	Dermatomal burning before rash	Early zoster can precede visible lesions
Pericarditis	Pleuritic, positional, better leaning forward	Myopericarditis, MI, tamponade

1.14 High-yield exam clues

Stem clue	Most likely diagnosis
Substernal pressure with exertion relieved by rest	Stable angina
Chest pain at rest with ST depression and positive troponin	NSTEMI
Chest pain at rest with normal troponin but ischemic ECG changes	Unstable angina
ST elevation in II, III, aVF	Inferior STEMI
Pleuritic chest pain, diffuse ST elevation, PR depression	Acute pericarditis
Sudden tearing pain radiating to back	Aortic dissection
Pleuritic pain, tachycardia, hypoxemia after surgery	Pulmonary embolism
Hypotension, JVD, muffled heart sounds	Cardiac tamponade
Sudden dyspnea, hypotension, unilateral absent breath sounds	Tension pneumothorax
Chest pain after forceful vomiting	Esophageal rupture
Chest pain after cocaine use	Coronary vasospasm/ACS

1.15 Common pitfalls

- Calling chest pain atypical too early.** Atypical does not mean safe. Older patients, diabetic patients, and women may present without classic crushing substernal pain.
- Using one negative troponin as a rule-out.** Troponin timing matters. Early presenters may need repeat testing.
- Forgetting dissection before anticoagulation.** A patient with tearing pain, pulse deficits, neurologic symptoms, or mediastinal widening needs dissection considered before routine ACS anticoagulation.
- Ignoring right ventricular MI.** Inferior MI with hypotension can involve the RV. These patients may be preload sensitive, and aggressive nitrates can worsen hypotension.
- Assuming reproducible tenderness excludes ACS.** It lowers probability but does not eliminate risk in a patient with concerning features.
- Treating all troponin elevations as plaque rupture.** Troponin elevation means myocardial injury. Sepsis, PE, renal disease, tachyarrhythmias, myocarditis, and heart failure can also elevate troponin.

1.16 Original diagram 2: ECG and troponin decision tree

Concerning chest pain


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      |
      v
12-lead ECG
      |
      +--- STEMI pattern? -----> Reperfusion pathway
      |
      +--- Ischemic ST depression/T inversion? --> ACS pathway + serial troponin
      |
      +--- Diffuse ST elevation/PR depression? -> Pericarditis/myopericarditis workup
      |
      +--- Normal or nondiagnostic? -----> Use story + serial troponin + risk score
                                           and screen for non-ACS emergencies

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Troponin trend:

Rising/falling + ischemic story = acute MI likely
 Positive but nonischemic context = myocardial injury; find the cause
 Negative too early = repeat if suspicion remains

1.17 Practice questions

1.17.1 Question 1

A 62-year-old man presents with 45 minutes of substernal pressure radiating to the left arm. He is diaphoretic. ECG shows ST elevation in leads II, III, and aVF. Initial troponin is pending.

What is the next best step?

Answer: Activate the STEMI reperfusion pathway. Do not wait for troponin. **Teaching point:** STEMI is diagnosed by ECG in the correct clinical context. Troponin may be negative early.

1.17.2 Question 2

A 54-year-old woman has chest pressure when climbing stairs. It resolves after 3 minutes of rest. Her ECG in clinic is normal. She has diabetes and hypertension.

What is the most likely diagnosis?

Answer: Stable angina due to chronic coronary syndrome. **Teaching point:** Predictable exertional symptoms relieved by rest suggest demand ischemia from fixed coronary disease.

1.17.3 Question 3

A 70-year-old man presents with sudden severe chest pain radiating to the back. Blood pressure is 190/105 in the right arm and 160/90 in the left arm. He has a new neurologic deficit.

What diagnosis must be prioritized?

Answer: Aortic dissection. **Teaching point:** Sudden tearing pain, pulse/BP differential, and neurologic findings are classic danger clues.

1.17.4 Question 4

A 29-year-old man has sharp chest pain that worsens when lying down and improves when leaning forward. ECG shows diffuse ST elevations and PR depressions.

What is the likely diagnosis?

Answer: Acute pericarditis. **Teaching point:** Positional pleuritic pain plus diffuse ECG changes points away from focal coronary-territory STEMI.

1.17.5 Question 5

A patient with pneumonia and septic shock has a mildly elevated troponin without ischemic ECG changes.

What is the best interpretation?

Answer: Myocardial injury from systemic illness or demand ischemia is possible; do not automatically label this type 1 MI. **Teaching point:** Troponin elevation must be interpreted in clinical context.

1.18 Chapter summary

Chest pain is approached by first identifying instability, then rapidly ruling out life-threatening causes. ECG and troponin are central, but neither replaces clinical reasoning. The most dangerous diagnoses include ACS/MI, aortic dissection, PE, tension pneumothorax, tamponade, and esophageal rupture. Stable angina is predictable exertional ischemia; ACS is new, worsening, rest, or prolonged ischemic pain; STEMI is an ECG-driven emergency requiring immediate reperfusion.

1.19 References and source notes

1. User-provided cardiology excerpt, *cardiology1.pdf*, pages 1-8. Used as a coverage checklist for stable angina, stress testing, cardiac catheterization, CAD treatment, and revascularization topics.
2. Gulati M, Levy PD, Mukherjee D, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain. *Circulation*. 2021. This guideline supports ECG-based evaluation, serial cardiac biomarker use, and structured risk assessment.
3. Merck Manual Professional Edition. Chest Pain; Some Causes of Chest Pain; Overview of Acute Coronary Syndromes. These references support the emergency chest pain differential, red flags, ECG/troponin/chest radiograph initial testing, and ACS triage framework.